GEOTWINAI

AI-Powered Digital Twin for Smart Cities

Nagpur Urban Planning & Decision Support Report

Study Area	Nagpur, Maharashtra
Satellite	Sentinel-2 Level-2A
Satellite Source	Microsoft Planetary Computer
Spatial Resolution	10 metres
Scene Date	2026-05-12
Generated By	GeoTwinAI Automated Report Generator

GeoTwinAI integrates geospatial infrastructure datasets, satellite remote sensing, environmental indices, flood analysis, heatmap analysis and machine learning to support smart-city planning and decision-making.

1. Executive Summary

GeoTwinAI is an AI-powered geospatial decision-support platform developed for urban planning and smart-city analysis in Nagpur, Maharashtra. The platform integrates infrastructure datasets, Sentinel-2 satellite observations, environmental indices, flood-risk analysis, heatmap analysis and machine-learning components. The selected Sentinel-2 scene has a reported cloud cover of 0.00% and a scene date of 2026-05-12. The environmental analysis includes NDVI, NDBI and NDWI. Infrastructure analysis includes buildings, hospitals, schools, parks, roads and water bodies. The platform additionally incorporates flood-risk and urban heatmap analysis to support environmental risk assessment. The final workflow connects GIS visualization, machine learning, automated data processing and Power BI analytics.

2. Project Objectives

• Develop an AI-powered Digital Twin framework for Nagpur.

• Integrate satellite and GIS datasets.

• Analyse vegetation and environmental conditions.

• Analyse built-up and water-related characteristics.

• Map important urban infrastructure.

• Analyse flood-risk conditions.

• Identify heat-related urban hotspots.

• Create machine-learning-based urban priority analysis.

• Provide interactive GIS visualization.

• Provide analytical dashboards through Power BI.

• Support data-driven urban planning decisions.

• Create an extensible framework for future data updates.

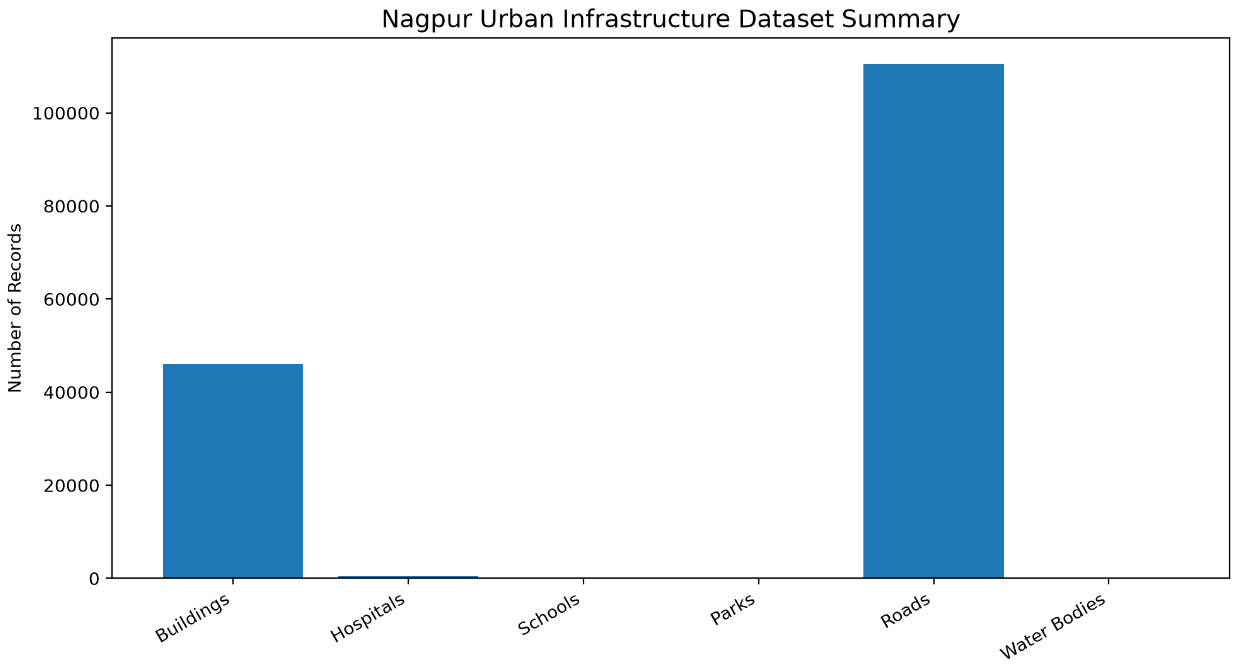
3. Data Sources

Dataset	Purpose
Buildings	Urban building and infrastructure analysis
Hospitals	Healthcare infrastructure mapping
Schools	Educational infrastructure mapping
Parks	Green and open-space analysis
Roads	Transportation network analysis
Water Bodies	Water-resource analysis
Sentinel-2	Satellite and environmental analysis
All Sentinel-2 Bands	Multi-spectral analysis
NDVI / NDBI / NDWI	Environmental and urban indicators
LULC	Land-use and land-cover analysis
Flood Risk	Flood vulnerability analysis

Heatmap	Urban hotspot analysis
Machine Learning	Urban priority prediction
Power BI	Interactive analytical dashboard

4. Urban Infrastructure Statistics

Infrastructure Type	Records
Buildings	46052
Hospitals	412
Schools	75
Parks	142
Roads	110566
Water Bodies	0

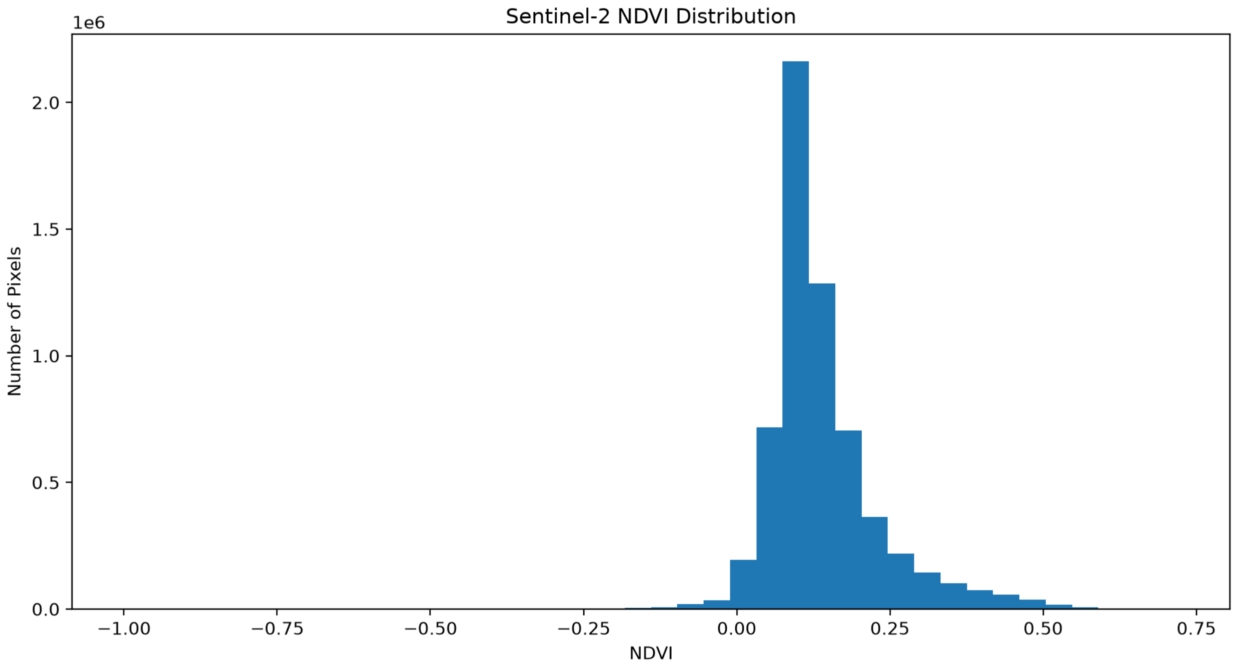


5. Sentinel-2 Satellite Analysis

The satellite analysis uses Sentinel-2 Level-2A imagery obtained through Microsoft Planetary Computer. The selected scene date is 2026-05-12, with a reported cloud cover of 0.00%. A common 10-metre spatial reference grid is used for integrated multi-band analysis. The processed Sentinel-2 dataset includes B01, B02, B03, B04, B05, B06, B07, B08, B8A, B09, B11 and B12.

6. Spectral Indices

Index	Purpose	Mean
NDVI	Vegetation condition	0.1392
NDBI	Built-up area indication	0.0804
NDWI	Water-related analysis	-0.1982



7. Sentinel-2 All-Band Analysis

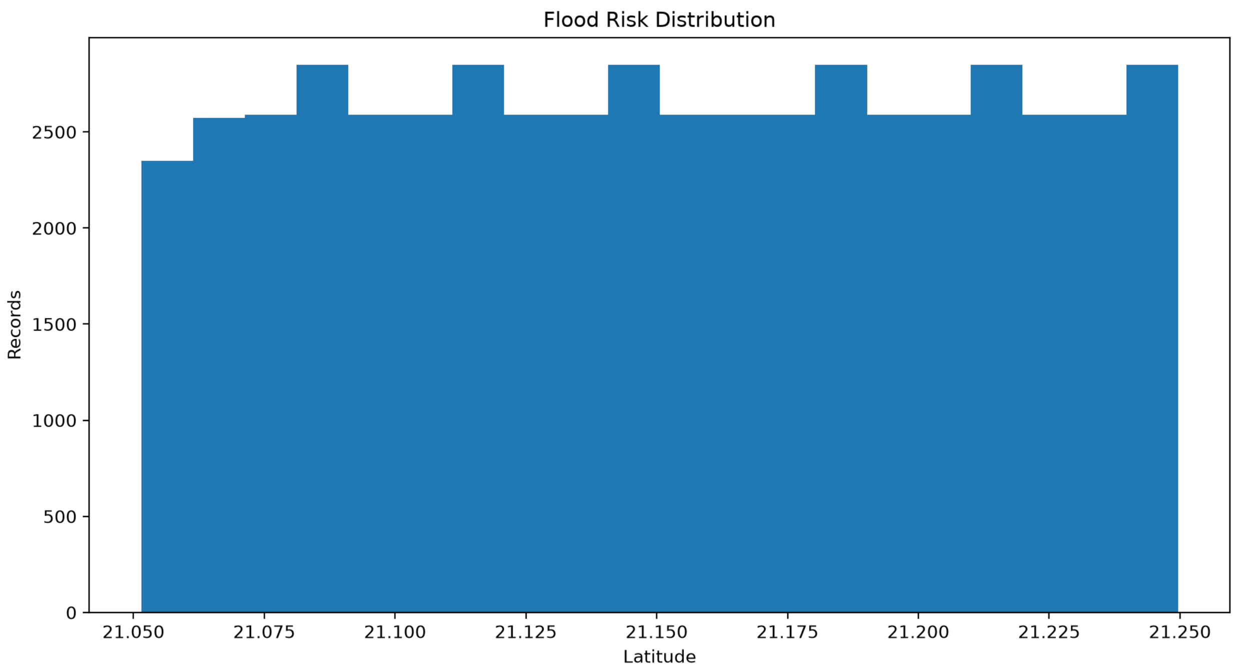
The all-band Sentinel-2 dataset preserves spectral information required for environmental and urban analysis. Processed bands include: B01 - Coastal/Aerosol B02 - Blue B03 - Green B04 - Red B05 - Vegetation Red Edge B06 - Vegetation Red Edge B07 - Vegetation Red Edge B08 - Near Infrared B8A - Narrow Near Infrared B09 - Water Vapour B11 - Short-Wave Infrared B12 - Short-Wave Infrared The combined spectral information supports vegetation, water, soil, built-up and land-cover analysis.

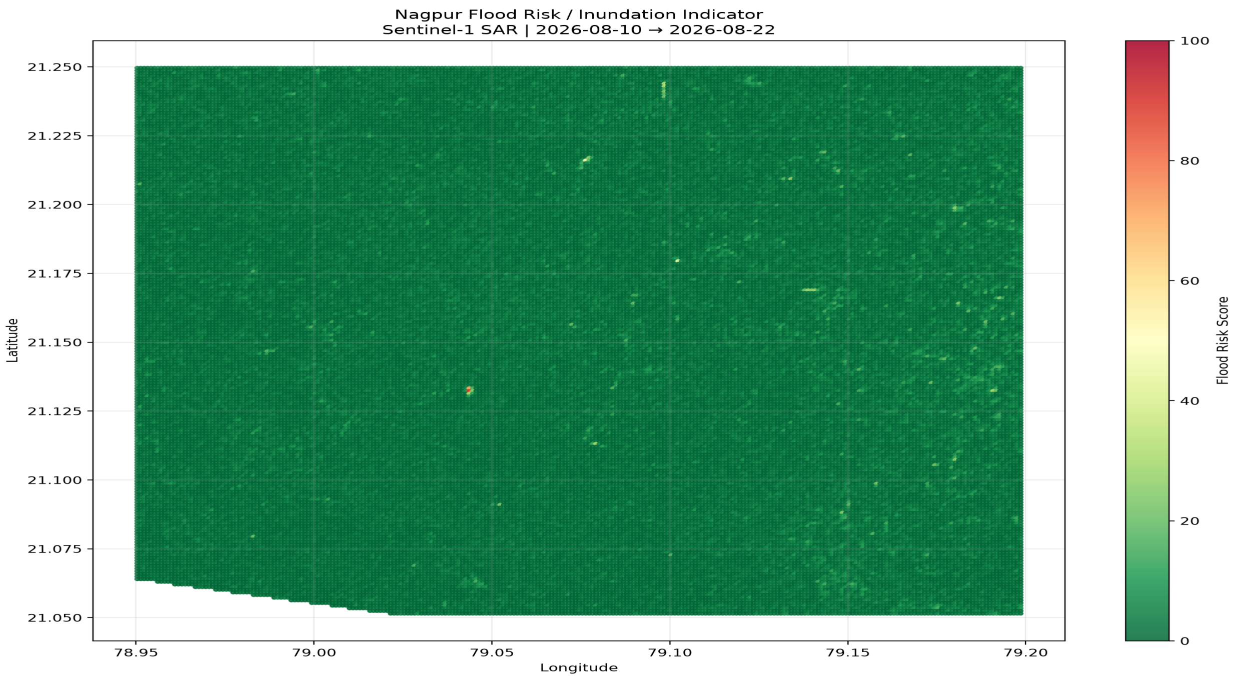
8. Land Use, Built-up Area and Green Cover

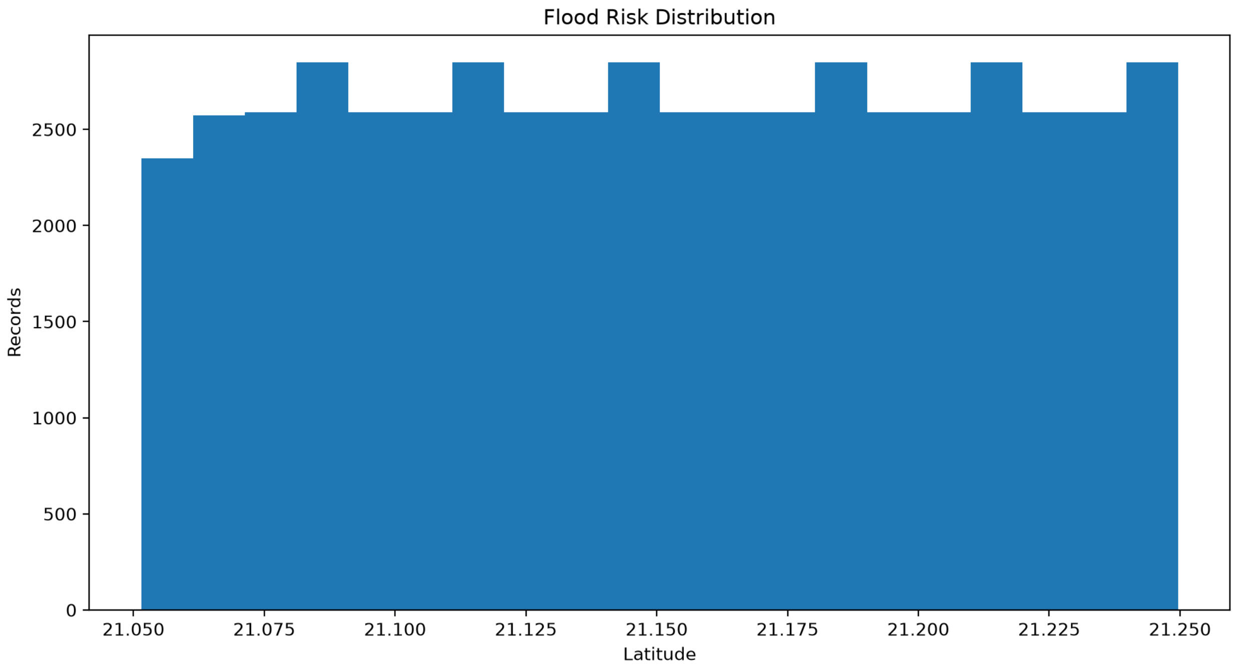
GeoTwinAI incorporates land-use and land-cover analysis to understand the spatial composition of the study area. Built-up analysis provides an indication of urbanized areas, while green-cover analysis supports identification of vegetation-rich and vegetation-deficient regions. LULC analysis can support planning decisions involving urban expansion, open spaces, vegetation conservation and land management.

9. Flood Risk Analysis

The GeoTwinAI system includes flood-risk analysis as an additional environmental risk component. The detected flood dataset contains approximately 53097 records. The flood analysis can be used to identify areas that may require additional attention for drainage planning, water management, emergency planning and urban resilience. The interactive GIS map can be used to spatially inspect the flood-risk information.

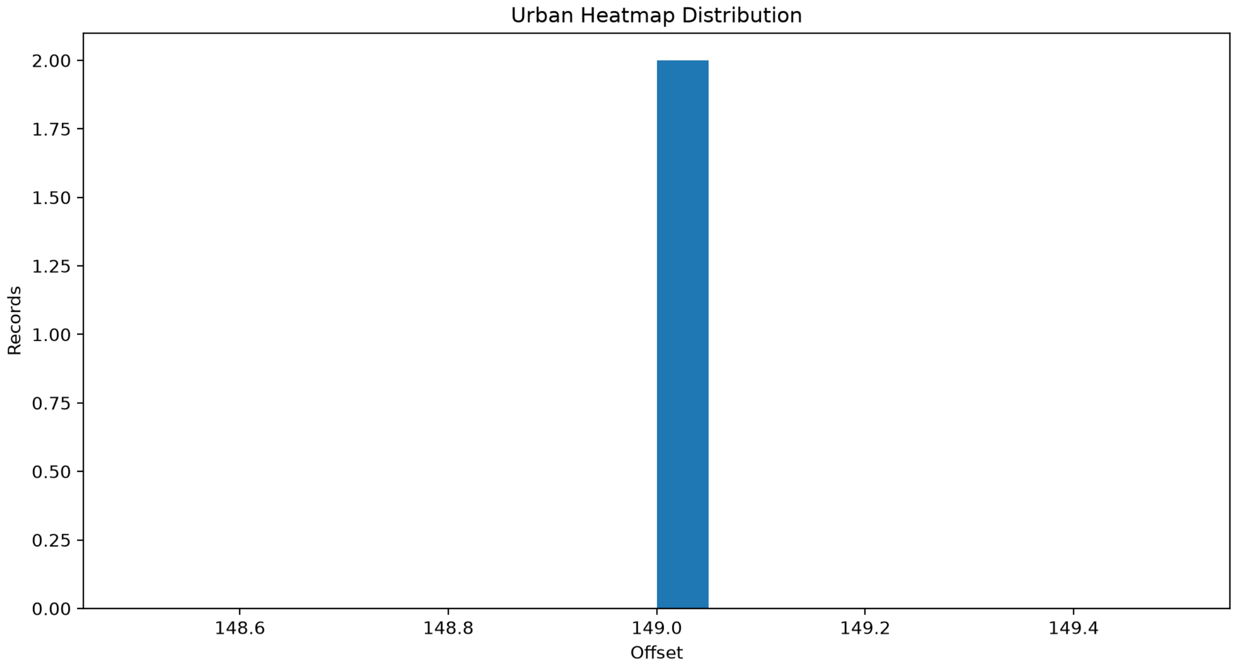


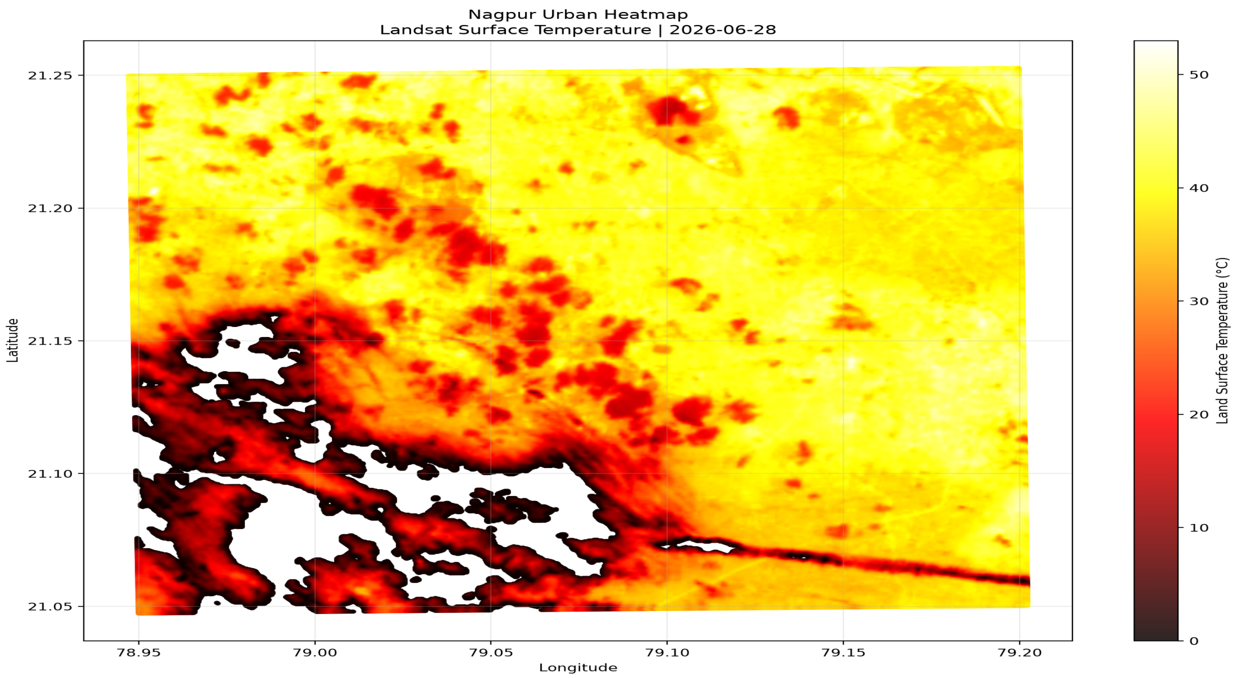


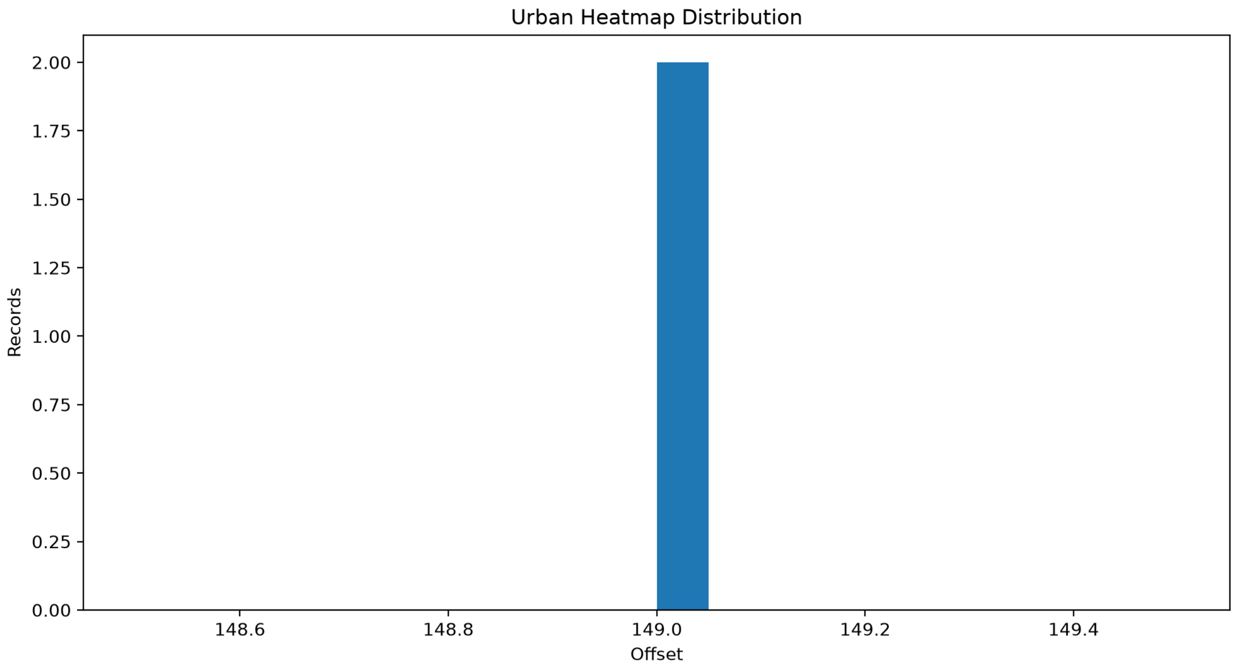

Flood Risk Distribution
Records
Latitude
0
500
1000
1500
2000
2500
21.050
21.075
21.100
21.125
21.150
21.175
21.200
21.225
21.250

10. Urban Heatmap Analysis

The GeoTwinAI platform also incorporates urban heatmap analysis to identify spatial concentrations or hotspots associated with heat-related conditions. The detected heatmap dataset contains approximately 5 records. Heatmap information can support urban heat mitigation, green-space planning, infrastructure planning and identification of areas requiring additional environmental attention.






Urban Heatmap Distribution
Records
Offset
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
148.6
148.8
149.0
149.2
149.4

11. Machine Learning Analysis

The machine-learning component of GeoTwinAI is designed to classify urban areas according to planning priority. Feature engineering combines relevant infrastructure, environmental and spatial attributes into a machine- learning dataset. The trained model can generate priority predictions such as High, Medium and Low. These predictions can support identification of areas requiring additional infrastructure or environmental attention. Model performance should be evaluated using accuracy, precision, recall and F1-score.

12. Interactive GIS Visualization

GeoTwinAI provides an interactive GIS visualization layer for exploring the spatial distribution of urban infrastructure and environmental-risk information. The interactive map is designed to allow users to switch layers on and off and inspect individual spatial records. Available layers include buildings, hospitals, schools, parks, roads, water bodies, flood-risk information and urban heatmap information when the corresponding datasets are available.

Interactive Map: Nagpur_Interactive_Map.html

13. GeoTwinAI Workflow

1. Data collection

2. Data cleaning

3. Satellite data acquisition

4. Satellite preprocessing

5. All-band processing

6. NDVI / NDBI / NDWI generation

7. LULC analysis

8. Built-up and green-cover analysis

9. Flood-risk analysis

10. Heatmap analysis

11. Feature engineering

12. Machine learning

13. Prediction

14. Interactive GIS visualization

15. Power BI dashboard

16. Urban planning decision support

17. Future automated data updates

14. Key Findings

• The infrastructure dataset contains 46052 building records.

• The healthcare dataset contains 412 hospital records.

• The education dataset contains 75 school records.

• The parks dataset contains 142 park records.

• The road dataset contains 110566 records.

• The water-body dataset contains 0 records.

• The selected Sentinel-2 scene has 0.00% reported cloud cover.

• The mean NDVI is 0.1392.

• The mean NDBI is 0.0804.

• The mean NDWI is -0.1982.

• Flood-risk records detected: 53097.

• Heatmap records detected: 5.

15. Recommendations

• Use the interactive GIS map to inspect spatial distribution of infrastructure.

• Use NDVI to identify vegetation-rich and vegetation-deficient areas.

• Use NDBI to identify highly built-up areas.

• Use NDWI and water-body information for water-resource planning.

• Use flood-risk analysis for drainage and urban-resilience planning.

• Use heatmap analysis to identify potential urban hotspots.

• Use ML priority classifications to identify areas requiring planning attention.

• Use Power BI for executive-level monitoring and comparative analysis.

• Use multi-date satellite observations to monitor changes over time.

• Integrate official MRSAC datasets wherever available for production-level analysis.

• Maintain year/date information so that future observations can be compared with historical data.

16. Limitations

• The Nagpur bounding box represents an analysis area and is not necessarily a formal administrative boundary.

• Satellite observations are affected by atmospheric and seasonal conditions.

• Cloud-cover metadata is scene-level and does not necessarily represent cloud conditions at every pixel.

• Flood and heatmap outputs depend on the underlying input datasets and processing methods.

- Machine-learning predictions depend on the quality and representativeness of the training dataset.
- Infrastructure records should be validated against authoritative datasets before operational deployment.
- Future-date information cannot be known until new observations or datasets become available.

17. Conclusion

GeoTwinAI provides a unified framework for combining geospatial infrastructure data, satellite remote sensing, environmental indices, flood-risk analysis, heatmap analysis and machine learning for smart-city planning. The integration of Sentinel-2 spectral information, environmental indicators and infrastructure datasets provides a spatially oriented understanding of urban conditions. The flood-risk and heatmap components extend the platform from infrastructure mapping to environmental-risk assessment. The interactive GIS map provides detailed spatial exploration, while Power BI can provide higher-level analytical dashboards for decision-makers. Future development can include multi-date satellite monitoring, improved administrative boundaries, official MRSAC datasets, advanced machine-learning models and automated acquisition of newly available observations. The overall system can therefore serve as a foundation for an AI-enabled urban planning and decision-support platform.

18. Technical Information

Programming	Python
Satellite	Sentinel-2
Product	Sentinel-2 Level-2A
Satellite Source	Microsoft Planetary Computer
Spatial Reference	10 metres
Indices	NDVI, NDBI, NDWI
Land Analysis	LULC / Built-up / Green Cover
Risk Analysis	Flood Risk / Heatmap
GIS	Folium Interactive HTML Map
Machine Learning	Urban Priority Classification
Dashboard	Microsoft Power BI
Report	Professional PDF